

CHAPTER 1: ALGEBRAIC METHODS

1.1 ARITHMETIC OPERATIONS WITH ALGEBRAIC FRACTIONS

1.1.1 Multiplication of Algebraic Fractions

- **Definition / Core Concept:** The process of combining two or more algebraic fractions into a single product by multiplying their respective numerators and denominators.
- **Governing Rule:**
 - **Cancel any common factors** between the numerators and the denominators first.
 - Multiply the remaining numerators together.
 - Multiply the remaining denominators together.
- **Conceptual Explanation:** Canceling common factors before multiplying prevents the resulting expression from becoming unnecessarily complex and ensures the final fraction is in its simplest form.
- **Assumptions and Constraints:**
 - Denominators must not be equal to zero: denominator $\neq 0$.
 - Factors must be identical to be canceled; terms within an expression cannot be canceled individually if they are part of a sum or difference.
- **Special Cases:**
 - **Zero Case:** If any numerator is 0, the entire product becomes 0 (provided no denominator is 0).
 - **Factorization Requirement:** Quadratic expressions (e.g., $x^2 - 1$) must often be factorized (e.g., $(x + 1)(x - 1)$) before common factors can be identified for cancellation.
- **Equations:**
 - Standard Form: $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$.
- **Practical Significance:** Essential for simplifying complex rational functions before differentiation or integration.

1.1.2 Division of Algebraic Fractions

- **Definition / Core Concept:** The process of dividing one algebraic fraction by another.
- **Governing Law:** To divide by a fraction, multiply the first fraction by the **reciprocal** of the second fraction.
- **Mechanistic Reasoning:** Division is the inverse of multiplication; multiplying by the reciprocal (the "flipped" fraction) achieves the same mathematical result as division.

- **Assumptions and Constraints:**

- The divisor fraction cannot be zero: $\frac{c}{d} \neq 0$ (implies $c \neq 0$ and $d \neq 0$).

- **Equations:**

- Standard Form: $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$.

- [Diagram: A visual flow showing the "flip" of the divisor fraction and the subsequent multiplication step].

1.1.3 Addition and Subtraction of Algebraic Fractions

- **Definition / Core Concept:** Combining fractions using additive or subtractive operators.

- **Governing Principle:** To add or subtract algebraic fractions, you must find a **common denominator**.

- **Mechanistic Steps:**

- Find the **Lowest Common Multiple (LCM)** of the existing denominators.
 - Adjust each fraction by multiplying the numerator and denominator by the factor required to reach the LCM.
 - Add or subtract the adjusted numerators over the common denominator.
 - Simplify the resulting numerator by expanding brackets and collecting like terms.

- **Special Cases:**

- **Boundary Scenario:** If denominators share no factors, the common denominator is the product of the two denominators: $\text{LCM} = (\text{denom}_1) \times (\text{denom}_2)$.
 - **Factorization Step:** If denominators are quadratic, factorize them first to find the simplest LCM.

- **Equations:**

- General Form: $\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$.

- **Significance:** Foundational for combining terms in algebraic proof and solving rational equations.

1.2 IMPROPER FRACTIONS

1.2.1 Definition and Notation

- **Definition:** An **improper algebraic fraction** is one where the degree of the numerator is **greater than or equal to** the degree of the denominator.

- **Notation (Degree of a Polynomial):** The **degree** of a polynomial is the largest exponent in the expression.

- Example: $x^3 + 5x - 9$ has degree 3.

- **Cause-Effect:** An improper fraction must be converted into a **mixed fraction** (a polynomial and a proper fraction) before it can be expressed in partial fractions.

1.2.2 Conversion to Mixed Fractions

- **Core Theory:** Any improper fraction can be rewritten as: $\frac{F(x)}{\text{divisor}} = Q(x) + \frac{R(x)}{\text{divisor}}$.
 - $F(x)$: Original numerator (dividend).
 - $Q(x)$: The quotient polynomial.
 - $R(x)$: The remainder.
- **Assumptions / Constraints:**
 - The degree of the remainder $R(x)$ must be **strictly less than** the degree of the divisor.
 - The divisor and remainder can be constants or functions of x .

Method 1: Algebraic Long Division

- **Process:** Traditional long division performed on polynomials until the remaining term has a lower degree than the divisor.
- **Mechanistic Reasoning:** Divides the higher-degree components into whole units (the quotient) and leaves the leftovers (the remainder) over the divisor.
- [Diagram: Labeled long division setup showing Dividend ($F(x)$), Divisor, Quotient ($Q(x)$), and Remainder (R)].

Method 2: The Identity Relationship

- **Governing Principle:** Use the relationship: $F(x) \equiv Q(x) \times \text{divisor} + \text{remainder}$.
- **Mechanistic Steps (Equating Coefficients):**
 - Multiply the mixed fraction form by the divisor to remove denominators.
 - This creates an **identity** (true for all values of x).
 - Find unknown constants by either:
 - **Substitution:** Choosing values of x that eliminate terms (usually the roots of the divisor).
 - **Comparing Coefficients:** Equating coefficients of corresponding powers of x on both sides of the identity.
- **Special Cases / Boundary Scenarios:**
 - **Dividing by Quadratic:** If the divisor is quadratic, the remainder is assumed to be linear: $Dx + E$.
 - **Dividing by Linear:** If the divisor is linear, the remainder is a constant: D .

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Multiplication:** $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$.
 - **Division:** $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$.
 - **Addition/Subtraction:** $\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$.
 - **Improper Fraction Condition:** degree of $F(x) \geq$ degree of divisor.
 - **Mixed Fraction Form:** $\frac{F(x)}{\text{divisor}} = Q(x) + \frac{R}{\text{divisor}}$.
 - **Conversion Identity:** $F(x) \equiv Q(x) \times \text{divisor} + \text{remainder}$.
 - **Constraint (Remainder Degree):** degree(R) < degree(divisor).
 - **Identity Symbol:** $\equiv$ (Means "equal for all values of x ").
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ULTRA-CONDENSED FINAL SUMMARY

- **Operations:** Cancel factors **before** multiplying; use reciprocal for division; find LCM for addition/subtraction.
- **Improper Fractions:** Occur when numerator degree $\geq$ denominator degree. Must convert for further manipulation.
- **Conversion:** Use **Long Division** or the identity $F(x) \equiv Q(x) \times \text{divisor} + R$.
- **Key Identity Property:** In $F(x) \equiv Q(x) \times \text{divisor} + R$, solve for constants by substituting roots of the divisor or comparing coefficients of x^n .
- **Remainder Rule:** If divisor is degree n , remainder is at most degree $n - 1$.

CHAPTER 2: FUNCTIONS AND GRAPH NOTES

2.1 THE MODULUS FUNCTION

- **Definition / Core Concept:** The **modulus** of a number a , written as $|a|$, is its **non-negative numerical value**. It is also known as the **absolute value**.
- **Governing Theory:** A modulus function is defined as:
 - $|f(x)| = f(x)$ when $f(x) \geq 0$

- $|f(x)| = -f(x)$ when $f(x) < 0$
- **Conceptual Explanation:** The modulus function acts like a pair of brackets; the value inside must be calculated first before applying the absolute value. It ensures the output is always ≥ 0 .
- **Assumptions and Constraints:**
 - $|f(x)| \geq 0$ for all real values of x .
- **Special Cases / Zero Conditions:**
 - $|f(x)| = 0$ only when $f(x) = 0$.
- **Equations and Procedures:**
 - **Graph Sketching ($y = |ax + b|$):**
 1. Sketch the line $y = ax + b$.
 2. Reflect the section of the graph below the x -axis in the x -axis.
 - **Solving Equations ($|f(x)| = g(x)$):** Solve the positive argument $f(x) = g(x)$ and the negative argument $-f(x) = g(x)$ separately.
- [Diagram: Graph showing $y = x$ and its reflection $y = |x|$ in the x -axis for $x < 0$].

2.2 FUNCTIONS AND MAPPINGS

- **Definition / Core Concept:** A **mapping** transforms one set of numbers into another. A mapping is a **function** if every input has exactly one distinct output.
- **Governing Principle:** Functions must be either **one-to-one** (each x has a unique y) or **many-to-one** (multiple x values can share the same y).
- **Conceptual Explanation:** If an input maps to more than one output (one-to-many), the mapping is **not** a function.
- **Assumptions and Limitations:**
 - **Domain:** The set of all possible inputs.
 - **Range:** The set of all possible outputs.
 - Mappings can be made into functions by restricting the domain (e.g., $y = \sqrt{x}$ is a function only if $x \geq 0$).
- **Boundary Scenarios:**
 - **Piecewise-defined functions:** Defined by different equations for different parts of the domain.
 - **Notation:** Functions can be written as $f(x) = \dots$ or $f : x \mapsto \dots$
- **Special Case (Identity Mapping):** In an identity, the range is identical to the domain.
- [Diagram: Three mapping diagrams: one-to-one (function), many-to-one (function), and one-to-many (not a function)].

2.3 COMPOSITE FUNCTIONS

- **Definition / Core Concept:** Combining two or more functions to create a new function.

- **Governing Law:** The notation $fg(x)$ means apply function g first, then apply function f to the result.
- **Mechanistic Reasoning:** $fg(x) = f(g(x))$. The output of g becomes the input for f .
- **Assumptions and Constraints:**
 - The **order** of combination is critical: $fg(x) \neq gf(x)$ in general.
 - The range of the inner function must lie within the domain of the outer function.
- **Special Notation:**
 - $f^2(x) = ff(x)$
- **Practical Significance:** Used to model multi-stage processes in mathematics and physics.
- [Diagram: A chain of two function machines showing $x \rightarrow g(x) \rightarrow f(g(x))$].

2.4 INVERSE FUNCTIONS

- **Definition / Core Concept:** An **inverse function** performs the opposite operation to the original function, mapping elements in the range back to the domain.
- **Governing Rule:** Inverse functions $f^{-1}(x)$ exist **only** for **one-to-one functions**.
- **Conceptual Explanation:**
 1. The domain of $f(x)$ is the range of $f^{-1}(x)$.
 2. The range of $f(x)$ is the domain of $f^{-1}(x)$.
- **Assumptions and Conditions:**
 1. $ff^{-1}(x) = f^{-1}f(x) = x$.
- **Geometric Relationship:** The graphs of $y = f(x)$ and $y = f^{-1}(x)$ are **reflections** of each other in the line $y = x$.
- **Equations and Procedures:**
 1. Set the function as $y = f(x)$.
 2. Rearrange the equation to make x the subject.
 3. Replace x with $f^{-1}(x)$ and y with x .
- **Special Case (Self-Inverse):** A function where $f(x) = f^{-1}(x)$; in this case, $ff(x) = x$.
- [Graph: $y = f(x)$ and $y = f^{-1}(x)$ shown as reflections across the dashed line $y = x$].

2.5 $y = |f(x)|$ AND $y = f(|x|)$

- **2.5.1 Graphing $y = |f(x)|$**
 - **Core Concept:** The modulus of the entire function output.
 - **Procedure:**
 1. Sketch $y = f(x)$.

2. Reflect any parts of the graph where $f(x) < 0$ (below the x -axis) in the x -axis.
 3. Delete the original parts below the x -axis.
- **2.5.2 Graphing $y = f(|x|)$**
 - **Core Concept:** The function of the absolute value of the input.
 - **Procedure:**
 1. Sketch $y = f(x)$ for $x \geq 0$.
 2. Reflect this part of the graph in the y -axis.
 - **Comparison:** $|f(x)|$ ensures no negative y -values; $f(|x|)$ creates symmetry about the y -axis.
 - [Graphs: Comparison of original $y = f(x)$, modulus output $y = |f(x)|$, and modulus input $y = f(|x|)$ for a quadratic curve].

2.6 COMBINING TRANSFORMATIONS

- **Definition / Core Concept:** Applying multiple changes (translations, stretches, reflections) to a base function $f(x)$.
- **Governing Transformations:**
 - **Translation (Vertical):** $f(x) + a$ by vector $\begin{pmatrix} 0 & a \end{pmatrix}$.
 - **Translation (Horizontal):** $f(x + a)$ by vector $\begin{pmatrix} -a & 0 \end{pmatrix}$.
 - **Stretch (Vertical):** $af(x)$ with scale factor a (parallel to y -axis).
 - **Stretch (Horizontal):** $f(ax)$ with scale factor $\frac{1}{a}$ (parallel to x -axis).
 - **Reflection (x-axis):** $-f(x)$ reflects in the x -axis.
 - **Reflection (y-axis):** $f(-x)$ reflects in the y -axis.
- **Assumptions and Mechanics:**
 - **Order is critical.** Standard sequence for $af(bx + c) + d$:
 1. Horizontal translation (C).
 2. Horizontal stretch (b).
 3. Vertical stretch (a).
 4. Vertical translation (d).

- [Graph: Multiple transformed curves based on a single base curve $y = f(x)$ with labeled points O, A, B].

2.7 SOLVING MODULUS PROBLEMS

- **Governing Theory:** Uses combinations of transformations and understanding of range/domain to solve equations and inequalities.
 - **Mechanistic Steps:**
 1. Sketch the modulus function using transformations (e.g., $y = a|x - b| + c$).
 2. Plot the second function (usually a line $y = mx + c$) on the same axes.
 3. Identify intersection points.
 4. Solve the resulting equations separately for the "original" and "reflected" parts of the modulus graph.
 - **Cause-Effect Reasoning:** Inequality solutions correspond to the intervals on the x -axis where one graph is above or below the other.
 - **Special Case (No Solutions):** Occurs if graphs do not intersect or if the modulus vertex is higher/lower than the required value.
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CONSOLIDATED EQUATION AND SYMBOL LIST

- **Modulus Definition:** $|x| = x$ if $x \geq 0$; $|x| = -x$ if $x < 0$
- **Identity Symbol:** $\equiv$
- **Mapping Notation:** $f : x \mapsto \dots$ or $f(x) = \dots$
- **Composition:** $fg(x) = f(g(x))$
- **Composition Power:** $f^2(x) = ff(x)$
- **Inverse Relationship:** $ff^{-1}(x) = f^{-1}f(x) = x$
- **Inverse Notation:** $f^{-1}(x)$
- **Domain/Range Inversion:** $\text{Domain}(f) = \text{Range}(f^{-1})$,
 $\text{Range}(f) = \text{Domain}(f^{-1})$
- **Vertical Translation:** $f(x) \rightarrow f(x) + a$
- **Horizontal Translation:** $f(x) \rightarrow f(x + a)$

- **Vertical Stretch:** $f(x) \rightarrow af(x)$
 - **Horizontal Stretch:** $f(x) \rightarrow f(ax)$
 - **Reflection (x-axis):** $y = -f(x)$
 - **Reflection (y-axis):** $y = f(-x)$
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ULTRA-CONDENSED FINAL SUMMARY

- **Modulus:** $|f(x)|$ is always ≥ 0 . Solve by considering positive and negative cases of the argument.
- **Functions:** Every input must have exactly one output. Inverse functions exist **only** for one-to-one functions.
- **Composition:** $fg(x)$ means apply g then f . Order matters.
- **Inverses:** Reflection of $f(x)$ in $y = x$. Range/Domain swap.
- **Modulus Graphs:** $|f(x)|$ reflects negative y to positive. $f(|x|)$ mirrors the positive x side to the negative x side.
- **Transformations:** Vertical changes are outside the brackets (affect y directly); horizontal changes are inside the brackets (affect x inversely).
- **Modulus Solving:** Always sketch graphs to identify the number of solutions and ensure solutions aren't "extra" (extraneous).

CHAPTER 3: TRIGONOMETRIC FUNCTIONS

3.1 SECANT, COSECANT AND COTANGENT

- **Definition / Core Concept:** These are known as the **reciprocal trigonometric functions**.
 - **Secant** ($\sec x$): The reciprocal of the cosine function.
 - **Cosecant** ($\csc x$): The reciprocal of the sine function.
 - **Cotangent** ($\cot x$): The reciprocal of the tangent function.

- **Governing Equations:**

- $\sec x = \frac{1}{\cos x}$
- $\operatorname{cosec} x = \frac{1}{\sin x}$
- $\cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}$

- **Conceptual Explanation:** These functions provide the ratios of sides in a right-angled triangle relative to the standard sine, cosine, and tangent ratios.

- **Assumptions, Conditions, and Limitations:**

- $\sec x$ is **undefined** for values of x for which $\cos x = 0$.
- $\operatorname{cosec} x$ is **undefined** for values of x for which $\sin x = 0$.
- $\cot x$ is **undefined** for values of x for which $\tan x = 0$ (or $\sin x = 0$).

- **Special Cases (Quadrant Signs):** The sign of the reciprocal function matches the sign of its corresponding primary function in each quadrant (CAST diagram).

- **1st Quadrant (All):** All positive.
- **2nd Quadrant (Sine):** $\operatorname{cosec} x$ is positive; others negative.
- **3rd Quadrant (Tan):** $\cot x$ is positive; others negative.
- **4th Quadrant (Cos):** $\sec x$ is positive; others negative.

- **Practical Significance:** Used to model oscillations and resonance in physical systems like bridges.

3.2 GRAPHS OF $\sec x$, $\operatorname{cosec} x$ AND $\cot x$

3.2.1 The Graph of $y = \sec x$

- **Definition:** The graphical representation of the secant function over a domain of real numbers.
- **Mechanistic Reasoning:** As $\cos x$ approaches 0, $\sec x$ approaches $\pm\infty$, creating vertical asymptotes.
- **Properties and Constraints:**
 - **Symmetry:** Symmetrical in the y -axis (even function).
 - **Period:** 360° or 2π radians.
 - **Vertical Asymptotes:** Occur at all values of x where $\cos x = 0$ (e.g., $\pm 90^\circ, \pm 270^\circ, \dots$ or any odd multiple of $\frac{\pi}{2}$).

- **Domain:** $x \in \mathbb{R} : x \neq 90^\circ, 270^\circ, 450^\circ, \dots$ or $x \in \mathbb{R} : x \neq \frac{(2n+1)\pi}{2}, n \in \mathbb{Z}$.
- **Range:** $y \leq -1$ or $y \geq 1$.
- [Graph: $y = \sec x$ showing vertical asymptotes at $x = \pm 90^\circ, \pm 270^\circ$ and local minima at $(0, 1)$ and local maxima at $(180^\circ, -1)$].

3.2.2 The Graph of $y = \operatorname{cosec} x$

- **Definition:** The graphical representation of the cosecant function.
- **Properties and Constraints:**
 - **Period:** 360° or 2π radians.
 - **Vertical Asymptotes:** Occur at all values of x where $\sin x = 0$ (e.g., $0^\circ, \pm 180^\circ, \pm 360^\circ, \dots$ or any multiple of π).
 - **Domain:** $x \in \mathbb{R} : x \neq 0^\circ, 180^\circ, 360^\circ, \dots$ or $x \in \mathbb{R} : x \neq n\pi, n \in \mathbb{Z}$.
 - **Range:** $y \leq -1$ or $y \geq 1$.
- [Graph: $y = \operatorname{cosec} x$ showing vertical asymptotes at $x = 0^\circ, \pm 180^\circ$ and local minima at $(90^\circ, 1)$ and local maxima at $(270^\circ, -1)$].

3.2.3 The Graph of $y = \cot x$

- **Definition:** The graphical representation of the cotangent function.
- **Properties and Constraints:**
 - **Period:** 180° or π radians.
 - **Vertical Asymptotes:** Occur at all values of x where $\tan x = 0$ (or $\sin x = 0$).
 - **Domain:** $x \in \mathbb{R} : x \neq 0^\circ, 180^\circ, 360^\circ, \dots$ or $x \in \mathbb{R} : x \neq n\pi, n \in \mathbb{Z}$.
 - **Range:** All real values ($y \in \mathbb{R}$).
- [Graph: $y = \cot x$ showing vertical asymptotes at multiples of 180° and x-intercepts at odd multiples of 90°].

3.3 USING $\sec x$, $\operatorname{cosec} x$ AND $\cot x$

- **Core Theory:** Solving equations and simplifying expressions involving reciprocal functions.
- **Governing Laws and Rules:**

- Always convert to $\sin x$, $\cos x$, or $\tan x$ as a first step for solving or simplifying.
 - **Limiting Condition:** Equations $\sec x = k$ and $\operatorname{cosec} x = k$ have **no solutions** if $-1 < k < 1$.
 - **Mechanistic Reasoning:** Because the ranges of $\sec x$ and $\operatorname{cosec} x$ are $|y| \geq 1$, their reciprocal functions $\cos x$ and $\sin x$ would require values $|y| > 1$, which is impossible.
 - **Substitution Rule:**
 - $\sec \theta = \frac{1}{\cos \theta}$
 - $\cot 2\theta = \frac{1}{\tan 2\theta}$
 - **Proving Identities:** Start with the more complex side (usually LHS) and use definitions to reach the other side.
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3.4 TRIGONOMETRIC IDENTITIES

- **Governing Principle:** New identities are derived from the fundamental identity $\sin^2 x + \cos^2 x \equiv 1$.
- **Derivations:**
 - **Identity 1:** Divide $\sin^2 x + \cos^2 x \equiv 1$ by $\cos^2 x$:

$$\frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} \equiv \frac{1}{\cos^2 x}$$
 - **Result:** $1 + \tan^2 x \equiv \sec^2 x$
 - **Identity 2:** Divide $\sin^2 x + \cos^2 x \equiv 1$ by $\sin^2 x$:

$$\frac{\sin^2 x}{\sin^2 x} + \frac{\cos^2 x}{\sin^2 x} \equiv \frac{1}{\sin^2 x}$$
 - **Result:** $1 + \cot^2 x \equiv \operatorname{cosec}^2 x$
- **Assumptions:** These identities hold for all values of x where the functions are defined.
- **Zero/Max/Min Conditions:**
 - $\sec^2 x$ is always ≥ 1 .

- $\operatorname{cosec}^2 x$ is always ≥ 1 .
 - **Practical Significance:** Essential for solving quadratic-form trigonometric equations by substituting for squared terms to create a single trigonometric ratio.
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3.5 INVERSE TRIGONOMETRIC FUNCTIONS

- **Core Concept:** The inverse functions of sine, cosine, and tangent.
- **Governing Rule:** To ensure inverse functions are **one-to-one**, the domains of the original functions must be restricted.

3.5.1 $y = \arcsin x$ (or $\sin^{-1} x$)

- **Definition:** The inverse of the restricted sine function.
- **Domain:** $-1 \leq x \leq 1$
- **Range:** $-\frac{\pi}{2} \leq \arcsin x \leq \frac{\pi}{2}$ (or $-90^\circ \leq y \leq 90^\circ$).
- [Graph: $y = \arcsin x$; a curve from $(-1, -\frac{\pi}{2})$ to $(1, \frac{\pi}{2})$ passing through the origin].

3.5.2 $y = \arccos x$ (or $\cos^{-1} x$)

- **Definition:** The inverse of the restricted cosine function.
- **Domain:** $-1 \leq x \leq 1$
- **Range:** $0 \leq \arccos x \leq \pi$ (or $0^\circ \leq y \leq 180^\circ$).
- [Graph: $y = \arccos x$; a decreasing curve from $(-1, \pi)$ through $(0, \frac{\pi}{2})$ to $(1, 0)$].

3.5.3 $y = \arctan x$ (or $\tan^{-1} x$)

- **Definition:** The inverse of the restricted tangent function.
- **Domain:** $x \in \mathbb{R}$ (all real values).
- **Range:** $-\frac{\pi}{2} < \arctan x < \frac{\pi}{2}$ (Note: strictly between; not equal).
- **Boundary Scenario:** Has horizontal asymptotes at $y = \pm \frac{\pi}{2}$.
- [Graph: $y = \arctan x$; a curve through the origin with horizontal asymptotes at $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$].

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Reciprocal Functions:**

- $\sec x = \frac{1}{\cos x}$
- $\operatorname{cosec} x = \frac{1}{\sin x}$
- $\cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}$

- **Pythagorean Identities:**

- $\sin^2 x + \cos^2 x \equiv 1$
- $1 + \tan^2 x \equiv \sec^2 x$
- $1 + \cot^2 x \equiv \operatorname{cosec}^2 x$

- **Inverse Functions (Domain $\rightarrow$ Range):**

- $\arcsin x : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- $\arccos x : [-1, 1] \rightarrow [0, \pi]$
- $\arctan x : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

- **Undefined Points:**

- $\sec x$ undefined at $x = \frac{(2n+1)\pi}{2}$
- $\operatorname{cosec} x$ undefined at $x = n\pi$
- $\cot x$ undefined at $x = n\pi$

- **Identity Symbol:** $\equiv$ (True for all values in the domain).

ULTRA-CONDENSED FINAL SUMMARY

- **Reciprocals:** Remember **Secant** goes with **Cosine**, **Cosecant** with **Sine**, **Cotangent** with **Tangent**.
- **Graphs:** $\sec$ and cosec live outside $(-1, 1)$. $\cot$ has same asymptotes as cosec .
- **New Identities:** Use $1 + \tan^2 x = \sec^2 x$ and $1 + \cot^2 x = \operatorname{cosec}^2 x$ to turn "mixed" quadratic equations into single-ratio equations.
- **Inverse Limits:** Range is key. $\arcsin$ and $\arctan$ are centered at 0; $\arccos$ is always positive $[0, \pi]$.

- **Solving:** Always check for undefined values (asymptotes) when solving equations. If $\cos x = 0$, $\sec x$ cannot exist.

CHAPTER 4: TRIGONOMETRIC ADDITION FORMULAE

4.1 ADDITION FORMULAE

- **Definition / Core Concept:** Formulas used to express the sine, cosine, and tangent of the sum or difference of two angles in terms of the trigonometric ratios of the individual angles. Also known as **compound-angle formulae**.
- **Governing Laws (The Addition Formulae):**
 - $\sin(A + B) \equiv \sin A \cos B + \cos A \sin B$
 - $\sin(A - B) \equiv \sin A \cos B - \cos A \sin B$
 - $\cos(A + B) \equiv \cos A \cos B - \sin A \sin B$
 - $\cos(A - B) \equiv \cos A \cos B + \sin A \sin B$
 - $\tan(A + B) \equiv \frac{\tan A + \tan B}{1 - \tan A \tan B}$
 - $\tan(A - B) \equiv \frac{\tan A - \tan B}{1 + \tan A \tan B}$
- **Derivation Logic (Geometric):**
 - Constructed using a right-angled triangle within another, where lengths are defined by trigonometric ratios (e.g., setting a hypotenuse to 1).
 - **Mechanistic Steps:**
 1. Identify angles α and β .
 2. Use perpendicular lines to create segments such as $\sin \beta$ and $\cos \beta$.
 3. Project these segments onto horizontal and vertical axes to find total lengths corresponding to $\sin(\alpha + \beta)$ and $\cos(\alpha + \beta)$.
- **Assumptions and Limitations:**
 - Tangent formulas are undefined if the denominator $1 \mp \tan A \tan B = 0$.
 - **Watch Out:** $\sin(A + B) \neq \sin A + \sin B$. A single counter-example (e.g., $A = 30^\circ, B = 60^\circ$) disproves this.

- **Special Cases:**
 - To prove difference formulae from addition formulae, substitute B with $-B$ and use the properties: $\cos(-B) = \cos B$ and $\sin(-B) = -\sin B$.
 - **Significance:** Allows the calculation of exact trigonometric values for angles that are sums/differences of standard angles (e.g., $15^\circ = 45^\circ - 30^\circ$).
 - [Diagram: Geometric construction on a coordinate plane with a unit hypotenuse $AE = 1$, angles α and β , and perpendicular segments EF , FC , BC , and CD used to derive sine and cosine addition].
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4.2 USING THE ANGLE ADDITION FORMULAE

- **Core Theory:** Application of addition formulae to find exact values and simplify expressions.
 - **Mechanistic Reasoning:**
 - Angles like 15° , 75° , 105° , 165° etc., can be broken down into combinations of 30° , 45° , 60° .
 - **Assumptions and Conditions:**
 - When given a ratio (e.g., $\sin A = -3/5$) and a quadrant constraint (e.g., $180^\circ < A < 270^\circ$), you must use $\sin^2 A + \cos^2 A = 1$ and a **CAST diagram** to determine the correct sign for the other ratios ($\cos A$, $\tan A$) before substituting into addition formulas.
 - **Equations:**
 - Example for exact value: $\sin 15^\circ = \sin(45^\circ - 30^\circ) = \frac{\sqrt{6} - \sqrt{2}}{4}$.
 - **Practical Significance:** High-precision engineering and physics problems where surd forms are required over decimal approximations.
-

4.3 DOUBLE-ANGLE FORMULAE

- **Definition / Core Concept:** Formulas expressing trigonometric ratios of $2A$ in terms of ratios of A .
- **Governing Laws (Derivations from Addition Formulae):**
 - **Sine:** Substitute $B = A$ into $\sin(A + B)$:
 - $\sin 2A \equiv 2 \sin A \cos A$
 - **Cosine:** Substitute $B = A$ into $\cos(A + B)$:

- $\cos 2A \equiv \cos^2 A - \sin^2 A$
 - Using $\cos^2 A + \sin^2 A = 1$, derived forms are:
 - $\cos 2A \equiv 2 \cos^2 A - 1$
 - $\cos 2A \equiv 1 - 2 \sin^2 A$
 - **Tangent:** Substitute $B = A$ into $\tan(A + B)$:
 - $\tan 2A \equiv \frac{2 \tan A}{1 - \tan^2 A}$
 - **Special Cases / Boundary Conditions:**
 - **Zero Case:** If $A = 0$, $\sin 2A = 0$, $\cos 2A = 1$, $\tan 2A = 0$.
 - **Undefined Case:** $\tan 2A$ is undefined if $\tan^2 A = 1$ (i.e., $A = 45^\circ, 135^\circ, \dots$).
 - **Significance:** Crucial for "eliminating" parameters in parametric equations and solving quadratic trigonometric equations.
-

4.4 SOLVING TRIGONOMETRIC EQUATIONS

- **Core Theory:** Using addition and double-angle identities to transform complex equations into a single trigonometric ratio that can be solved within a given interval.
 - **Mechanistic Reasoning:**
 - Substitute identities (e.g., replace $\cos 2x$ with a form in $\sin x$ or $\cos x$ to match other terms).
 - Factorize the resulting equation (often a quadratic).
 - Find principal values using a calculator.
 - Use a **CAST diagram** or graph to find all solutions in the required range.
 - **Assumptions / Constraints:**
 - If the angle is 2θ or 3θ , the range of the solution must be adjusted accordingly (e.g., if $0 \leq \theta < 2\pi$, then $0 \leq 3\theta < 6\pi$).
 - **Limiting Conditions:** Equations involving $\cos 2A = k$ must have $|k| \leq 1$.
 - **Special Case (Triple Angles):** $\sin 3A$ can be solved by treating it as $\sin(2A + A)$ and expanding.
-

4.5 SIMPLIFYING $a \cos x \pm b \sin x$

- **Core Theory:** Any expression of the form $a \cos x + b \sin x$ can be written as a single sine or cosine function with a phase shift.
- **Governing Laws (The Harmonic Form):**
 - $a \sin x \pm b \cos x \equiv R \sin(x \pm \alpha)$

- $a \cos x \pm b \sin x \equiv R \cos(x \mp \alpha)$
 - **Mechanistic Calculations:**
 - **Constant R :** $R = \sqrt{a^2 + b^2}$ (where $R > 0$).
 - **Angle α :** Found by equating coefficients. Usually:
 - $R \cos \alpha = a$ and $R \sin \alpha = b$.
 - $\tan \alpha = \frac{b}{a}$ (where $0 < \alpha < 90^\circ$ or $\pi/2$).
 - **Zero/Max/Min Behavior:**
 - **Maximum Value:** The maximum value of $R \cos(x \pm \alpha)$ is R .
 - **Minimum Value:** The minimum value is $-R$.
 - **Occurrences:** Maximum occurs when the argument of the cosine function is $0, 360^\circ, \dots$.
 - **Significance:** Vital for modeling periodic phenomena (e.g., microwaves, sound waves) and finding stationary points without calculus.
-

4.6 PROVING TRIGONOMETRIC IDENTITIES

- **Definition / Core Concept:** Demonstrating that two trigonometric expressions are identical for all values of the variable.
 - **Governing Law:** One side of the identity (usually the more complex one) is manipulated until it is identical to the other side.
 - **Mechanistic Reasoning:**
 - **Substitution:** Replace terms using addition, double-angle, or reciprocal identities.
 - **Fraction Handling:** Find common denominators to merge terms.
 - **Common Factors:** Dividing numerator and denominator by a common term (e.g., dividing by $\cos^2 A$ to produce $\tan A$ terms).
 - **Assumptions:** The identity is only valid where all involved functions (especially $\tan, \sec, \operatorname{cosec}, \cot$) are defined.
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Addition Formulae:**
 - $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

- $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
 - $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$
 - **Double-Angle Formulae:**
 - $\sin 2A = 2 \sin A \cos A$
 - $\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$
 - $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
 - **Harmonic Form Constants:**
 - $R = \sqrt{a^2 + b^2}$
 - $\alpha = \arctan\left(\frac{b}{a}\right)$
 - **Range Constraints:**
 - $R > 0$
 - $0 < \alpha < 90^\circ$ (unless otherwise specified)
 - **Symbols:**
 - $\equiv$: Identity (equal for all values)
 - $\pm/\mp$: Plus-minus / Minus-plus (indicates sign reversal in cosine formulae)
-

ULTRA-CONDENSED FINAL SUMMARY

- **Addition:** Sine maintains signs ($+$ stays $+$); Cosine reverses signs ($+$ becomes $-$).
- **Double Angles:** Use the version of $\cos 2A$ that matches the other terms in your equation ($\sin^2$ vs $\cos^2$) to solve quadratics.
- **Harmonic Form (R, α):** Essential for finding Max (R) and Min ($-R$) of combined wave functions. Solve

$$\tan \alpha = \frac{\text{coeff of term not in } R \text{ expansion}}{\text{coeff of term in } R \text{ expansion}}.$$

- **Solving:** Always adjust the search interval if the angle is a multiple (e.g., 2θ).
- **Proving:** Start with the more complex side; convert to $\sin$ and $\cos$ if stuck.

CHAPTER 5: EXPONENTIALS AND LOGARITHMS

5.1 EXPONENTIAL FUNCTIONS

- **Definition / Core Concept:** A function of the form $f(x) = a^x$, where a is a positive constant.
- **Notation:** In the expression a^x , x is defined as the **index**, **power**, or **exponent**.
- **Governing Theory (Graph Shapes):**
 - The graph of $y = a^x$ is a smooth curve passing through $(0, 1)$ for all positive a because $a^0 = 1$.
 - **Increasing Case ($a > 1$):** The function is increasing. As x increases, a^x grows without limit. As x decreases, a^x tends toward 0 but never reaches it.
 - **Decreasing Case ($0 < a < 1$):** The function is decreasing. As x increases, a^x tends toward 0.
- **Conceptual Explanation:** Exponential growth occurs because the rate of change is proportional to the current value.
- **Assumptions and Constraints:**
 - Base a must be positive ($a > 0$).
 - If $a = 1$, the function is a horizontal line $y = 1$.
- **Special Cases and Zero Conditions:**
 - **Asymptote:** The x -axis ($y = 0$) is a horizontal asymptote for the curve.
 - **Reflections:** The graph of $y = (1/a)^x$ (or $y = a^{-x}$) is a reflection of $y = a^x$ in the y -axis.
- **Equations:**
 - General form: $y = a^x$
 - Value at origin: $f(0) = 1$
- **Significance:** Used to model rapidly changing quantities such as population, compound interest, or radioactive decay.

- [Graph: Three curves on one axis showing $y = 3^x$, $y = 2^x$, and $y = 1.5^x$, all intersecting at $(0, 1)$].
-

5.2 $y = e^{ax+b} + c$

- **Definition / Core Concept:** The natural exponential function based on Euler's number e .
- **The Constant e :** An irrational mathematical constant approximately equal to 2.71828.
- **Governing Law (Differentiation):**
 - The gradient function of $f(x) = e^x$ is exactly the same as the function itself.
 - $\frac{d}{dx}(e^x) = e^x$
 - $\frac{d}{dx}(e^{kx}) = ke^{kx}$
- **Mechanistic Reasoning:** As the base a in a^x increases, the gradient of the curve at any point also increases. The value e is the unique base where the gradient at any point equals the y -coordinate at that point.
- **Graphing $y = e^{ax+b} + c$:**
 - **Horizontal Asymptote:** Occurs at $y = c$.
 - **y-intercept:** Found by setting $x = 0$, giving $(0, e^b + c)$.
- **Transformations:**
 - e^{x+b} : Translation by vector $(-b \ 0)$.
 - e^{ax} : Horizontal stretch with scale factor $1/a$.
 - $+c$: Vertical translation by vector $(0 \ c)$.
- **Equations:**
 - Derivative of e^{kx} : $f'(x) = ke^{kx}$
 - General Transformation Form: $y = Ae^{kx} + c$
- [Graph: $y = e^x$ shown with its asymptote at $y = 0$ and intercept at $(0, 1)$].

5.3 NATURAL LOGARITHMS

- **Definition / Core Concept:** The inverse function of the exponential function $y = e^x$. Written as $\ln x$.
- **Notation:** $\ln x \equiv \log_e x$.
- **Governing Theory (Inverse Relationship):**
 - The graph of $y = \ln x$ is a reflection of $y = e^x$ in the line $y = x$.
- **Conceptual Explanation:** Taking the natural logarithm is the inverse operation of raising e to a power.
- **Assumptions and Constraints:**
 - **Domain:** $\ln x$ is defined **only** for $x > 0$.
 - **Range:** $\ln x$ can take any real value ($-\infty < y < \infty$).
- **Special Cases / Zero Conditions:**
 - **x-intercept:** $\ln(1) = 0$ (passes through $(1, 0)$).
 - **Asymptote:** The y -axis ($x = 0$) is a vertical asymptote.
 - **Values of 1:** $\ln(e) = 1$.
- **Equations:**
 - Inverse Property 1: $e^{\ln x} = x$ (for $x > 0$).
 - Inverse Property 2: $\ln(e^x) = x$ (for all real x).
- **Significance:** Essential for solving equations where the unknown is in the exponent (e.g., $e^{2x} = 7$).
- [Graph: $y = e^x$ and $y = \ln x$ shown as reflections across the line $y = x$].

5.4 LOGARITHMS AND NON-LINEAR DATA

- **Definition / Core Concept:** Using logarithms to transform non-linear relationships into linear ones ($Y = MX + C$) to identify constants.

5.4.1 Case 1: $y = ax^n$ (Power Law)

- **Mechanistic Logic:** Taking logs of both sides:
 - $\log y = \log(ax^n)$

- $\log y = \log a + n \log x$
- **Linear Comparison:**
 - $Y = \log y$
 - $X = \log x$
 - **Gradient (M):** n
 - **Intercept (C):** $\log a$
- **Identification:** A graph of $\log y$ against $\log x$ yielding a straight line indicates a relationship of form $y = ax^n$.

5.4.2 Case 2: $y = ab^x$ (Exponential Law)

- **Mechanistic Logic:** Taking logs of both sides:
 - $\log y = \log(ab^x)$
 - $\log y = \log a + x \log b$
 - **Linear Comparison:**
 - $Y = \log y$
 - $X = x$
 - **Gradient (M):** $\log b$
 - **Intercept (C):** $\log a$
 - **Identification:** A graph of $\log y$ against x yielding a straight line indicates a relationship of form $y = ab^x$.
 - **Assumptions / Constraints:**
 - Common logs ($\log_{10}$) or natural logs ($\ln$) can be used; the relationship remains linear.
 - **Zero/Max/Min Case:**
 - If the gradient n (Case 1) or $\log b$ (Case 2) is negative, the original function is decreasing.
 - **Practical Significance:** Allows scientists to determine unknown constants in physical laws (e.g., Zipf's law or Kleiber's law) from experimental data.
 - [Table: Comparisons showing Y, X, M, C for both $y = ax^n$ and $y = ab^x$].
-

5.5 EXPONENTIAL MODELLING

- **Definition / Core Concept:** Using functions of form $P = Ae^{kt}$ to model real-world scenarios.
 - **Governing Principle:** The rate of change of a quantity is proportional to the quantity itself.
 - $\frac{dP}{dt} \propto P \implies \frac{dP}{dt} = kP$
 - **Mechanistic Reasoning:**
 - **Growth** ($k > 0$): Rate of increase is proportional to size (e.g., population growth).
 - **Decay** ($k < 0$): Rate of decrease is proportional to size (e.g., radioactive decay).
 - **Definitions of Constants:**
 - A : The **initial value** of the quantity (at $t = 0$).
 - k : The rate constant or growth/decay factor.
 - **Limitations of Models:**
 - Models may become invalid for very large values of t (e.g., population cannot grow forever due to resource limits).
 - **Equations:**
 - Model form: $y = Ae^{kt}$
 - Rate of change: $\frac{dy}{dt} = kAe^{kt} = ky$
 - **Significance:** Critical for predicting future states in biology (infection spread), physics (half-life), and finance.
 - [Graph: Exponential decay curve showing P decreasing toward an asymptote].
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Euler's Number:** $e \approx 2.71828$
- **Exponential Derivative:** $\frac{d}{dx}(e^{kx}) = ke^{kx}$
- **Natural Log Definition:** $\ln x = \log_e x$
- **Inverse Identities:** $e^{\ln x} = x$ and $\ln(e^x) = x$
- **Log Laws (Transformation Case 1):** $y = ax^n \implies \log y = n \log x + \log a$
- **Log Laws (Transformation Case 2):** $y = ab^x \implies \log y = x \log b + \log a$
- **Linear Form:** $Y = MX + C$

- **Modelling Form:** $P = Ae^{kt}$
 - **Modelling Rate:** $\frac{dP}{dt} = kP$
 - **Identity Symbol:** $\equiv$ (True for all values in domain)
-

ULTRA-CONDENSED FINAL SUMMARY

- **Exponentials:** $y = a^x$ always passes $(0, 1)$. e^x is its own derivative.
- **Logs:** $\ln x$ is the inverse of e^x . Domain is $x > 0$. Reflection in $y = x$.
- **Linearizing Data:**
 - Graph $\log y$ vs $\log x$ to find power n in $y = ax^n$.
 - Graph $\log y$ vs x to find base b in $y = ab^x$.
- **Modelling:** A is the initial value ($t = 0$). k is the growth constant. Rate is always $k \times$ (current value).
- **Equations:** Always use natural logs ($\ln$) to isolate variables in exponents of base e .

CHAPTER 6: DIFFERENTIATION

6.1 DIFFERENTIATING $\sin x$ AND $\cos x$

- **Definition / Core Concept:** Determining the rate of change of primary trigonometric functions.
- **Governing Laws (Standard Results):**
 - $\frac{d}{dx}(\sin x) = \cos x$
 - $\frac{d}{dx}(\cos x) = -\sin x$
- **Conceptual and Mechanistic Explanation:**
 - Derived from **first principles** using small angle approximations as $h \rightarrow 0$:
 - $\sin h \approx h$

$$\blacksquare \quad \cos h \approx 1 - \frac{1}{2}h^2$$

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0.$$

- **Assumptions, Conditions, and Limitations:**

- **Global Requirement:** You MUST use **radians** when differentiating trigonometric functions.

- **Equations (Chain Rule Variants):**

- $\frac{d}{dx}(\sin kx) = k \cos kx$
- $\frac{d}{dx}(\cos kx) = -k \sin kx$

- **Special Cases / Zero Behavior:**

- **Stationary Points:** Occur where $\frac{dy}{dx} = 0$. For $y = \sin x$, stationary points are at odd multiples of $\frac{\pi}{2}$.

- **Practical Significance:** Modeling velocity and acceleration in oscillatory motion (e.g., displacement of a tennis ball).

- [Graph: Comparison of $y = \sin x$ and its derivative $y' = \cos x$ showing alignment of stationary points and intercepts].

6.2 DIFFERENTIATING EXPONENTIALS AND LOGARITHMS

- **Definition / Core Concept:** Rates of change for functions with base e , base a , and natural logarithms.

- **Governing Laws:**

- **Exponential (e):** $\frac{d}{dx}(e^x) = e^x$
- **Natural Log:** $\frac{d}{dx}(\ln x) = \frac{1}{x}$

- **Mechanistic Explanation:**
 - The value e (≈ 2.71828) is the unique base where the gradient of the function is exactly equal to the y -coordinate at any point.
 - $\ln x$ is the inverse of e^x . Its derivative is the reciprocal of the input.
 - **Assumptions and Constraints:**
 - For $\ln x$, the domain must be $x > 0$.
 - For $\ln kx$, the derivative is still $\frac{1}{x}$ because $\ln kx = \ln k + \ln x$ and the constant $\ln k$ differentiates to 0.
 - **Equations (Generalized Forms):**
 - $\frac{d}{dx}(e^{kx}) = ke^{kx}$
 - **Base a Rule:** $\frac{d}{dx}(a^x) = a^x \ln a$
 - **General Base a :** $\frac{d}{dx}(a^{kx}) = ak^{kx} \ln a$
 - **Practical Significance:** Essential for modeling population growth and radioactive decay where rates are proportional to size.
-

6.3 THE CHAIN RULE

- **Definition / Core Concept:** A rule for differentiating **composite functions** (functions of functions).
- **Governing Law:**

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$
- **Mechanistic Reasoning:**
 - Break the function into an "outer" function $y = f(u)$ and an "inner" function $u = g(x)$.
 - The total rate of change is the product of the rate of change of the outer function with respect to the inner, and the inner with respect to x .
- **Derived Forms (Function Notation):**

- **Power of a function:** If $y = (f(x))^n$, then

$$\frac{dy}{dx} = n(f(x))^{n-1} f'(x)$$
 - **Function of a function:** If $y = f(g(x))$, then

$$\frac{dy}{dx} = f'(g(x))g'(x)$$
 - **Special Case (Inverse Rates):**

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$$
 - Used when it is easier to express x in terms of y (e.g., finding the gradient of $x = \sin 2y$).
 - [Diagram: A flowchart showing $x \rightarrow [g(x)] \rightarrow u \rightarrow [f(u)] \rightarrow y$ and the corresponding derivative products].
-

6.4 THE PRODUCT RULE

- **Definition / Core Concept:** Differentiating an expression where two functions of x are **multiplied** together.
- **Governing Law:**

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$
 - If $y = uv$, then
- **Function Notation:**
 - If $f(x) = g(x)h(x)$, then

$$f'(x) = g(x)h'(x) + h(x)g'(x)$$
- **Mechanistic Steps:**
 - Identify u and v .
 - Differentiate each to find $\frac{du}{dx}$ and $\frac{dv}{dx}$.
 - Apply the formula and simplify.
- **Assumptions and Watch Outs:**

- Do NOT differentiate each part separately and multiply them (

$$\frac{dy}{dx} \neq \frac{du}{dx} \times \frac{dv}{dx}.$$

- Significance:** Used for functions like $y = x^2 \sin x$ or $y = e^x \ln x$.
-

6.5 THE QUOTIENT RULE

- Definition / Core Concept:** Differentiating an expression where one function of x is divided by another.

- Governing Law:**

$$\text{If } y = \frac{u}{v}, \text{ then } \frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}.$$

- Function Notation:**

$$\text{If } f(x) = \frac{g(x)}{h(x)}, \text{ then } f'(x) = \frac{h(x)g'(x) - g(x)h'(x)}{(h(x))^2}.$$

- Mechanistic Reasoning:**

- The denominator is always squared.
- The order of terms in the numerator is critical** due to the minus sign: always

$\frac{du}{dx}$

start with the denominator (v) multiplied by the derivative of the numerator ($\frac{du}{dx}$).

- Assumptions and Limitations:**

- The divisor function must be non-zero: $v(x) \neq 0$.

- Special Cases:**

- Stationary Points:** If $\frac{dy}{dx} = 0$, then the numerator of the quotient rule result must equal zero: $v \frac{du}{dx} - u \frac{dv}{dx} = 0$.
-

6.6 DIFFERENTIATING TRIGONOMETRIC FUNCTIONS

- **Core Theory:** Using Chain, Product, and Quotient rules to find derivatives of reciprocal and inverse trigonometric functions.

- **Governing Laws (Reciprocal Functions):**

- $\frac{d}{dx}(\tan kx) = k \sec^2 kx$
- $\frac{d}{dx}(\sec kx) = k \sec kx \tan kx$
- $\frac{d}{dx}(\operatorname{cosec} kx) = -k \operatorname{cosec} kx \cot kx$
- $\frac{d}{dx}(\cot kx) = -k \operatorname{cosec}^2 kx$

- **Governing Laws (Inverse Functions):**

- $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$

- **Mechanistic Reasoning (Derivations):**

$$\frac{\sin x}{\cos x}$$

- $\tan x$ is derived via Quotient Rule on $\frac{\sin x}{\cos x}$.

- $\arcsin x$ is derived by setting $x = \sin y$, finding $\frac{dx}{dy} = \cos y$, and

$$\text{using } \frac{dy}{dx} = \frac{1}{\cos y} \text{ combined with } \cos y = \sqrt{1 - \sin^2 y}.$$

- **Boundary Conditions / Domains:**

- For $\arcsin x$ and $\arccos x$, the derivative is **undefined** at $x = \pm 1$ (vertical tangent).
 - For $\arctan x$, the derivative is valid for all real x .
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Trigonometric:**

- $\frac{d}{dx}(\sin kx) = k \cos kx$
- $\frac{d}{dx}(\cos kx) = -k \sin kx$
- $\frac{d}{dx}(\tan kx) = k \sec^2 kx$
- $\frac{d}{dx}(\sec kx) = k \sec kx \tan kx$
- $\frac{d}{dx}(\operatorname{cosec} kx) = -k \operatorname{cosec} kx \cot kx$
- $\frac{d}{dx}(\cot kx) = -k \operatorname{cosec}^2 kx$

- **Exponential & Logarithmic:**

- $\frac{d}{dx}(e^{kx}) = ke^{kx}$
- $\frac{d}{dx}(a^{kx}) = ka^{kx} \ln a$
- $\frac{d}{dx}(\ln x) = \frac{1}{x}$

- **Inverse Trigonometric:**

- $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$

- **Core Rules:**

- **Chain:** $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$
 - **Product:** $(uv)' = uv' + vu'$
 - **Quotient:** $\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$
 - **Reciprocal:** $\frac{dy}{dx} = \left(\frac{dx}{dy}\right)^{-1}$
-

ULTRA-CONDENSED FINAL SUMMARY

- **Rule of Signs:** Derivatives of "Co-" functions (cos, cot, cosec, arccos) always start with a **negative** sign.
- **Chain Rule:** Differentiate the "outside" function first, then multiply by the derivative of the "inside" function.
- **Product vs Quotient:** Product is a **sum**; Quotient is a **difference** (starting with u) divided by v^2 .

- **Log Shortcut:** The derivative of $\ln(f(x))$ is always $\frac{f'(x)}{f(x)}$ (via Chain Rule).
- **Angles:** Ensure calculator is in **Radians**; small angle approximations $\sin h \approx h$ underpin all trig calculus derivations.
- **Exponentials:** e^x is the only non-zero function that is its own derivative. For other bases, you must multiply by $\ln(\text{base})$.

CHAPTER 7: INTEGRATION

7.1 INTEGRATING STANDARD FUNCTIONS

- **Definition / Core Concept:** Integration is the inverse process of differentiation. It involves finding a function whose derivative is the given integrand.
- **Governing Laws (Standard Integrals):**

$$\circ \int x^n dx = \frac{x^{n+1}}{n+1} + c \quad (\text{Condition: } n \neq -1)$$

$$\circ \int e^x dx = e^x + c$$

$$\circ \int \frac{1}{x} dx = \ln |x| + c$$

$$\circ \int \cos x dx = \sin x + c$$

$$\circ \int \sin x dx = -\cos x + c$$

$$\circ \int \sec^2 x dx = \tan x + c$$

$$\circ \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c$$

$$\circ \int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$\circ \int \sec x \tan x dx = \sec x + c$$

- **Conceptual Explanation:** Because the derivative of $\sin x$ is $\cos x$, the integral of $\cos x$ must be $\sin x$. The constant $+C$ is necessary for indefinite integrals because differentiating a constant yields zero, meaning many functions share the same derivative.
- **Assumptions and Limitations:**
 - **Trigonometry:** When applying limits to integrated trigonometric functions, you MUST work in **radians**.
 - **Logarithms:** The modulus sign in $\ln |x|$ is used to remove difficulties when evaluating integrals for negative values of x .
- **Special Cases:**

- **Zero/Undefined:** The power rule for x^n is undefined for $n = -1$; this specific case is handled by the natural logarithm.
 - **Practical Significance:** Used in archaeology for carbon dating based on exponential decay and in physics to find the area under a curve.
-

7.2 INTEGRATING $f(ax + b)$

- **Core Concept:** Integrating a function where the argument is a linear transformation of x .

- **Governing Law:** If $\int f'(x)dx = f(x) + c$, then:

$$\int f'(ax + b)dx = \frac{1}{a}f(ax + b) + c$$

- **Mechanistic Reasoning:** This is the **reverse of the chain rule**. When differentiating $f(ax + b)$, you multiply by the derivative of the inner function (a). Therefore, when integrating, you must divide by that same constant a to reach the original function.

- **Assumptions and Constraints:**

- This method ONLY works when the argument is **linear** (form $ax + b$).
- **Watch Out:** It cannot be used for functions like $\cos(2x^2 + 3)$.

- **Derived Forms:**

- $\int e^{ax+b}dx = \frac{1}{a}e^{ax+b} + c$

- $\int \frac{1}{ax + b}dx = \frac{1}{a} \ln |ax + b| + c$

- $\int \sin(ax + b)dx = -\frac{1}{a} \cos(ax + b) + c$

7.3 USING TRIGONOMETRIC IDENTITIES

- **Definition / Core Concept:** Substituting trigonometric expressions with equivalent identities to transform them into standard integrable forms.
 - **Governing Theory:** Expressions that cannot be integrated directly (like squared terms) are replaced by identical expressions that can be integrated.
 - **Key Identity Substitutions:**
 - **Integrating $\tan^2 x$:** Use $1 + \tan^2 x \equiv \sec^2 x$ to get

$$\int (\sec^2 x - 1) dx$$
 - **Integrating $\cot^2 x$:** Use $1 + \cot^2 x \equiv \operatorname{cosec}^2 x$ to get

$$\int (\operatorname{cosec}^2 x - 1) dx$$
 - **Integrating $\sin^2 x$:** Use the double angle formula
 $\cos 2x \equiv 1 - 2 \sin^2 x$ to substitute

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$
 - **Integrating $\cos^2 x$:** Use $\cos 2x \equiv 2 \cos^2 x - 1$ to substitute

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$
 - **Integrating Products:** Use $\sin 2A \equiv 2 \sin A \cos A$ to substitute for

$$\sin 3x \cos 3x = \frac{1}{2} \sin 6x$$

terms like
 - **Mechanistic Reasoning:** Products and powers are reduced to sums of first-degree functions of multiple angles, which are standard integrals.
-

7.4 REVERSE CHAIN RULE

- **Definition / Core Concept:** Integrating functions that are recognizable as the result of differentiating a composite function.
- **Governing Laws (Functional Forms):**
 - **Form 1 (Logarithmic):**
$$\int k \frac{f'(x)}{f(x)} dx = k \ln |f(x)| + c$$

- **Form 2 (Power):**

$$\int k f'(x)(f(x))^n dx = \frac{k}{n+1} (f(x))^{n+1} + c$$

- **Mechanistic Procedure:**

$$k \frac{f'(x)}{f(x)}$$

- Observe if the integrand is in the form $k f'(x)(f(x))^n$ or $\frac{k f'(x)}{f(x)}$.
- Try integrating by suggesting a solution like $\ln |f(x)|$ or $(f(x))^{n+1}$.
- Differentiate your suggestion to check it against the original integrand.
- Adjust any constant factors as necessary.

- **Assumptions and Watch Outs:**

- **Constraint:** You cannot use this method if the numerator does not contain a

derivative of the denominator. For example, $\int \frac{1}{x^2 + 3} dx$ is **not** $\ln |x^2 + 3|$ because there is no x term in the numerator.

- **Special Case:**

$$\frac{\sin x}{\cos x}$$

- Integration of functions like $\tan x$: Rewrite as $\frac{\sin x}{\cos x}$ and use Form 1 (with a negative sign adjustment).

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Standard Algebraic:** $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$)

- **Standard Exponential:** $\int e^x dx = e^x + c$

- **Standard Reciprocal:** $\int \frac{1}{x} dx = \ln |x| + c$

- **Standard Trig (Sine/Cosine):**

- $\int \cos x dx = \sin x + c$

- $\int \sin x dx = -\cos x + c$
 - **Standard Trig (Derived):**
 - $\int \sec^2 x dx = \tan x + c$
 - $\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c$
 - $\int \operatorname{cosec}^2 x dx = -\cot x + c$
 - $\int \sec x \tan x dx = \sec x + c$
 - **Linear Form:** $\int f'(ax + b) dx = \frac{1}{a} f(ax + b) + c$
 - **Reverse Chain (Fraction):** $\int k \frac{f'(x)}{f(x)} dx = k \ln |f(x)| + c$
 - **Reverse Chain (Power):** $\int k f'(x) (f(x))^n dx = \frac{k}{n+1} (f(x))^{n+1} + c$
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ULTRA-CONDENSED FINAL SUMMARY

- **Standard Integrals:** Memorize the 9 core results; always include $+c$ for indefinite integrals.
- **Linear Arguments:** For $f(ax + b)$, integrate as normal and divide by the coefficient a .
- **Trig Identities:** Use for powers ($\sin^2$, $\cos^2$, $\tan^2$) to convert them into first-degree multiple angles or integrable squared terms ($\sec^2$).
- **Reverse Chain Rule:** If the integrand is "derivative $\times$ function", the integral is based on the function raised to a power or $\ln$ of the function. Always verify by differentiating the result.
- **Radians:** Mandatory for all trigonometric evaluations involving limits.
- **Logarithm Rule:** Use $\ln |x|$ to ensure the argument is positive.

CHAPTER 8: NUMERICAL METHODS

8.1 LOCATING ROOTS

- **Definition / Core Concept:** A root of a function $f(x)$ is a value of x for which $f(x) = 0$.
- **Governing Principle (Sign Change Rule):** If the function $f(x)$ is **continuous** on the interval $[a, b]$ and $f(a)$ and $f(b)$ have **opposite signs**, then $f(x)$ has at least one root x such that $a < x < b$.
- **Conceptual and Mechanistic Explanation:**
 - The graph of $y = f(x)$ crosses the x -axis whenever $f(x) = 0$.
 - For a continuous curve to move from a positive y -value to a negative y -value (or vice versa), it must pass through $y = 0$.
- **Assumptions and Constraints:**
 - **Continuity:** The function must not "jump" (no holes or vertical asymptotes) within the interval $[a, b]$.
 - **Radians:** When locating roots of trigonometric functions, the calculator and all calculations must be in **radians**.
- **Special Cases and Boundary Scenarios:**
 - **Multiple Roots (Odd Number):** There may be an odd number of roots within the interval $[a, b]$; a sign change will still occur.
 - **Multiple Roots (Even Number):** If there is an even number of roots (or a "touching" point), a sign change might **not** occur, even though roots exist.
 - **Vertical Asymptotes:** A sign change may occur across a vertical asymptote (e.g., $y = \tan x$); in this case, there is **no root** at the sign change.
- **Equations and Procedures:**
 - Root Identification: $f(x) = 0$.
 - Interval verification: Check if $f(a) \times f(b) < 0$.
 - Refining accuracy: To prove a root α is correct to n decimal places, show a sign change occurs between the upper and lower bounds of that value (e.g., for 1.763, check $f(1.7625)$ and $f(1.7635)$).
- **Significance:** Many mathematical models (e.g., Moon/Sun positions) cannot be solved with ordinary algebraic equations and require numerical approximation.
- [Graph: A curve crossing the x-axis between points a and b where $f(a) > 0$ and $f(b) < 0$].
- [Graph: Multiple roots within an interval showing an odd number of crossings].
- [Graph: Multiple roots within an interval showing an even number of roots with no net sign change].
- [Graph: A vertical asymptote where the function changes from $+\infty$ to $-\infty$ without crossing the x-axis].

8.2 FIXED POINT ITERATION

- **Definition / Core Concept:** A method used to find a value of x for which $f(x) = 0$ by repeatedly applying a formula.
 - **Governing Law:**
 - Rearrange $f(x) = 0$ into the form $x = g(x)$.
 - Use the iterative formula: $x_{n+1} = g(x_n)$.
 - **Mechanistic Reasoning:**
 - The root of $f(x) = 0$ is the x-coordinate of the intersection between the line $y = x$ and the curve $y = g(x)$.
 - Starting with an initial value x_0 , each iteration "maps" the result back onto the curve until the values converge on the intersection point.
 - **Convergence Types:**
 - **Successive Iterations:** Values get closer to the root from the same direction.
 - **Staircase Diagram:** Successive iterations create a "staircase" shape approaching the root.
 - **Cobweb Diagram:** Successive iterations alternate above and below the root, creating a "spiral" or "cobweb" shape.
 - **Assumptions and Limitations:**
 - **Convergence:** Not all starting values x_0 or rearrangements $g(x)$ will converge to a root.
 - **Divergence:** If iterations move further away from the root, the sequence is **divergent**.
 - **Validity:** The rearrangement $x = g(x)$ is often only valid for specific domains (e.g., $x \neq 0$).
 - **Equations:**
 - Iteration Formula: $x_{n+1} = g(x_n)$
 - Target Identity: $f(x) \equiv x - g(x) = 0$
 - **Practical Significance:** Essential for computer algorithms and engineering software to solve non-linear equations to high precision.
 - [Diagram: A staircase diagram showing x_0, x_1, x_2 converging on the intersection of $y = x$ and $y = g(x)$ from one side].
 - [Diagram: A cobweb diagram showing iterations "spiralling" into the root from alternating sides of the intersection].
 - [Diagram: A divergent iteration where the cobweb or staircase moves away from the intersection point].
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CONSOLIDATED EQUATION AND SYMBOL LIST

- **Root Condition:** $f(x) = 0$
 - **Sign Change Identity:** $f(a) < 0, f(b) > 0 \implies \text{root} \in (a, b)$ (for continuous f)
 - **Fixed Point Form:** $x = g(x)$
 - **Iterative Sequence:** $x_{n+1} = g(x_n)$
 - **Initial Approximation:** x_0
 - **Convergence Goal:** $x_{n+1} \approx x_n \approx \alpha$ (where α is the root)
 - **Upper/Lower Bounds:** ± 0.5 of the last decimal place (e.g., $1.763 \rightarrow [1.7625, 1.7635]$)
 - **Symbols:**
 - α : The exact root.
 - n : The iteration number.
 - $\in$: Member of an interval.
 - $\mathbb{R}$: Real numbers.
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ULTRA-CONDENSED FINAL SUMMARY

- **Locating Roots:** Look for a sign change (+ to - or vice versa) on a **continuous** interval.
- **Failures:** Even roots don't change signs; vertical asymptotes change signs without a root.
- **Precision:** To prove a root to d decimal places, test the sign at the boundaries of that precision level (± 0.5 of the smallest unit).
- **Iteration:** Rearrange $f(x) = 0 \rightarrow x = g(x)$. Different rearrangements result in different convergence/divergence behaviors.
- **Visualizing Iteration:** **Staircase** (one-sided approach) or **Cobweb** (alternating approach). If the graph "spirals" out, it is **divergent**.
- **Radians:** Essential for all numerical methods involving trigonometric functions.